

Deliverable #3

SE 3A04: Software Design II – Large System Design

Tutorial Number: T03

Group Number: G1

Group Members:

- Nadeem Elsayed
- Muhammad Ullah
- Kaitlyn Kenwell
- Aiden Sanvido
- Kurlan Beeharry

IMPORTANT NOTES

- You do NOT need to provide a text explanation of each diagram; the diagram should speak for itself
- Please document any non-standard notations that you may have used
 - Rule of Thumb: if you feel there is any doubt surrounding the meaning of your notations, document them
- Some diagrams may be difficult to fit into one page
 - It is OK if the text is small but please ensure that it is readable when printed
 - If you need to break a diagram onto multiple pages, please adopt a system of doing so and thoroughly explain how it can be reconnected from one page to the next; if you are unsure about this, please ask me
- Please submit the latest version of Deliverable 1 and Deliverable 2 with Deliverable 3
 - They do not have to be a freshly printed versions; the latest marked versions are OK
- If you do NOT have a Division of Labour sheet, your deliverable will NOT be marked

1 Introduction

1.1 Purpose

This document provides further information about the **SCEMAS** system architecture, including state chart diagrams, sequence diagrams, and a detailed class diagram.

This document will provide visibility and documentation for **SCEMAS**, an environmental monitoring and alert system. This document will discuss the purpose of the system, the scope of the application, user characteristics, product requirements, and use case diagrams.

1.2 System Description

An overview of the system description can be found in deliverable 2. This document acts as an extension of deliverable 2, providing more context in form of state charts, sequence diagrams, and a detailed class diagram.

1.3 Overview

This document is organized by chart/diagram type. Section 2 contains relevant state chart diagrams, section 3 contains relevant sequence diagrams, and section 4 provides a detailed UML diagram of the system

2 State Charts for Controller Classes

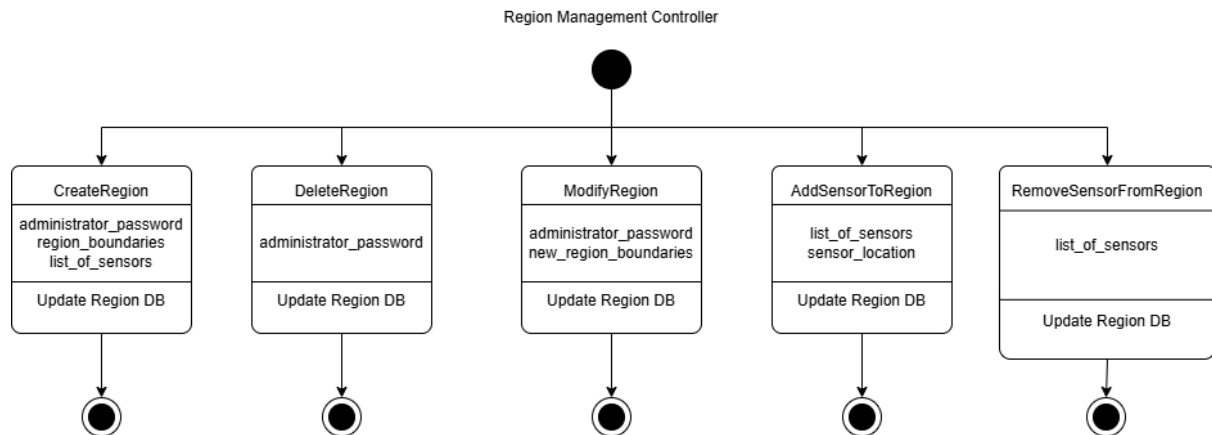


Figure 2.1. Region Management Controller State Diagram

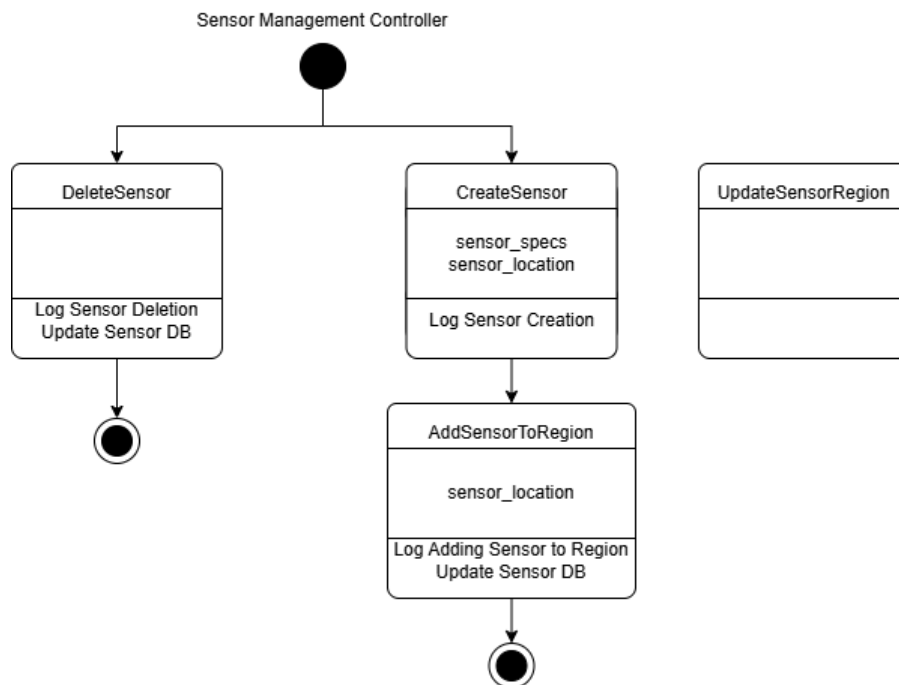


Figure 2.2. Sensor Management Controller State Diagram

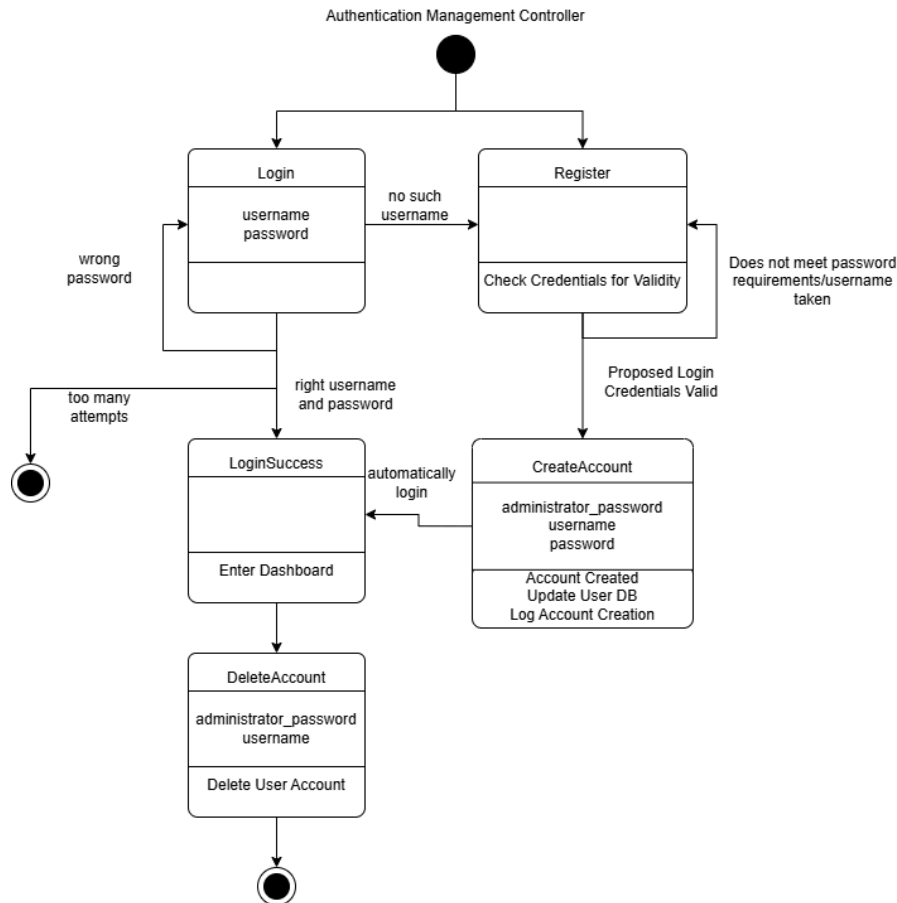


Figure 2.3. Authentication Management Controller State Diagram

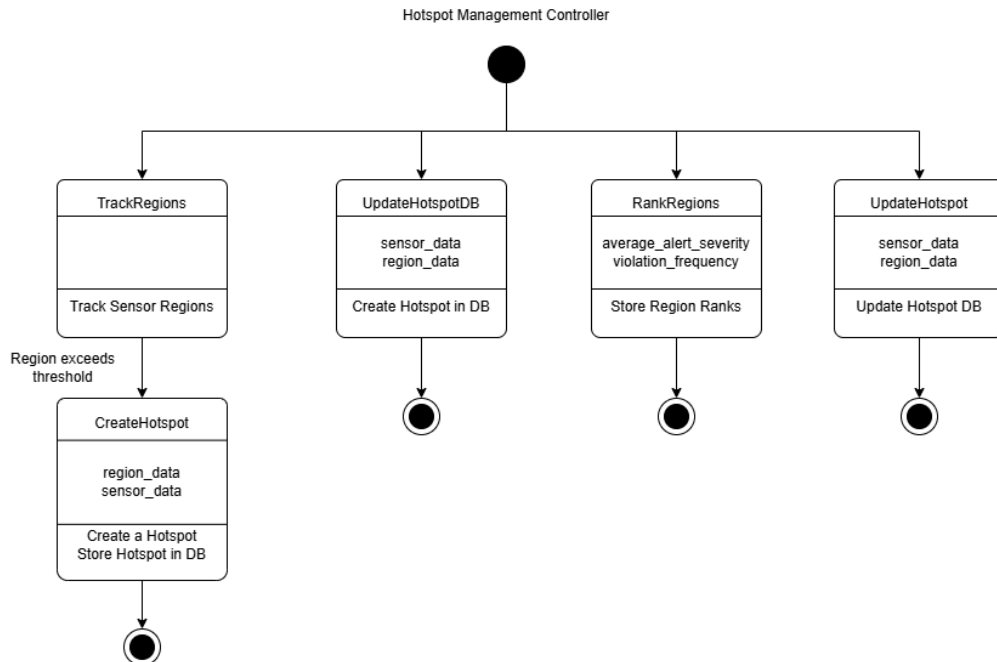


Figure 2.4. Hotspot Management Controller State Diagram

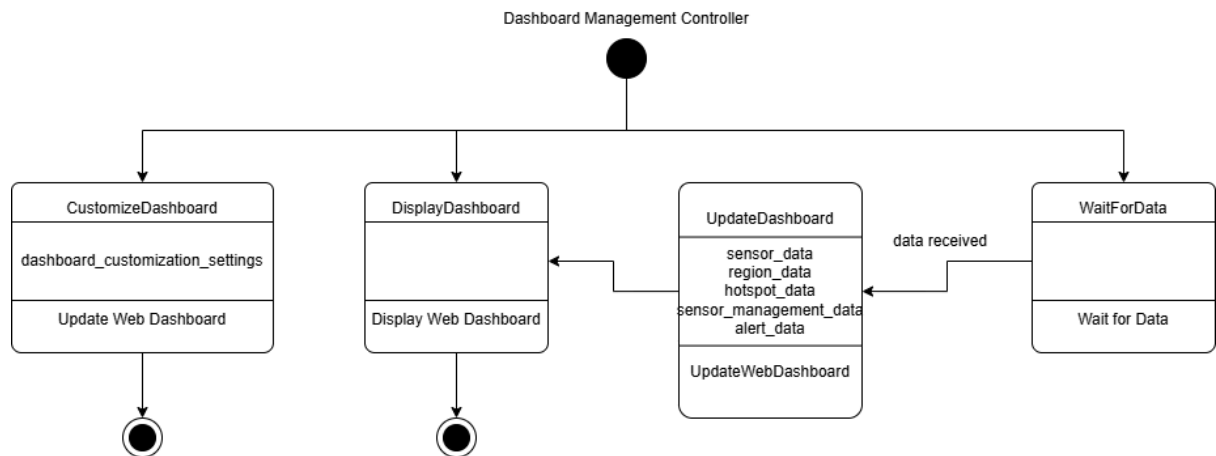


Figure 2.5. Dashboard Management Controller State Diagram

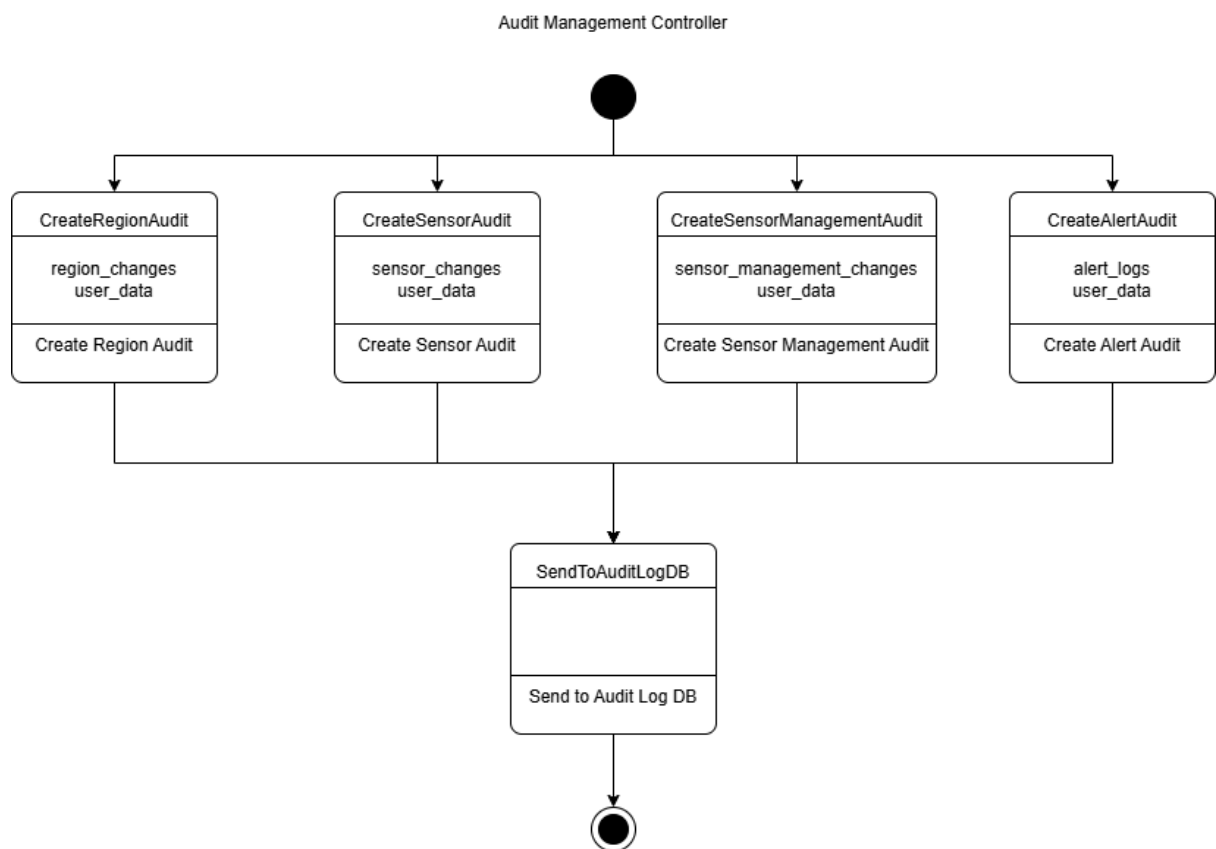


Figure 2.6. Audit Management Controller State Diagram

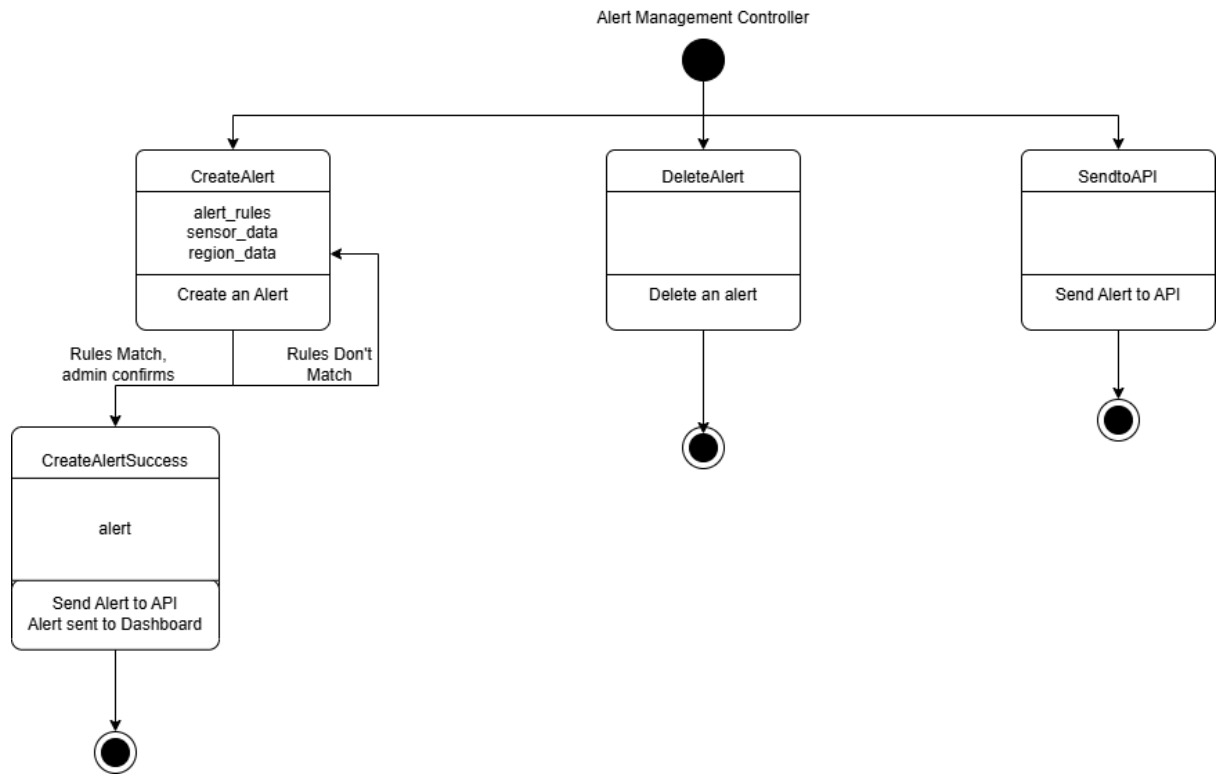


Figure 2.7. Alert Management Controller State Diagram

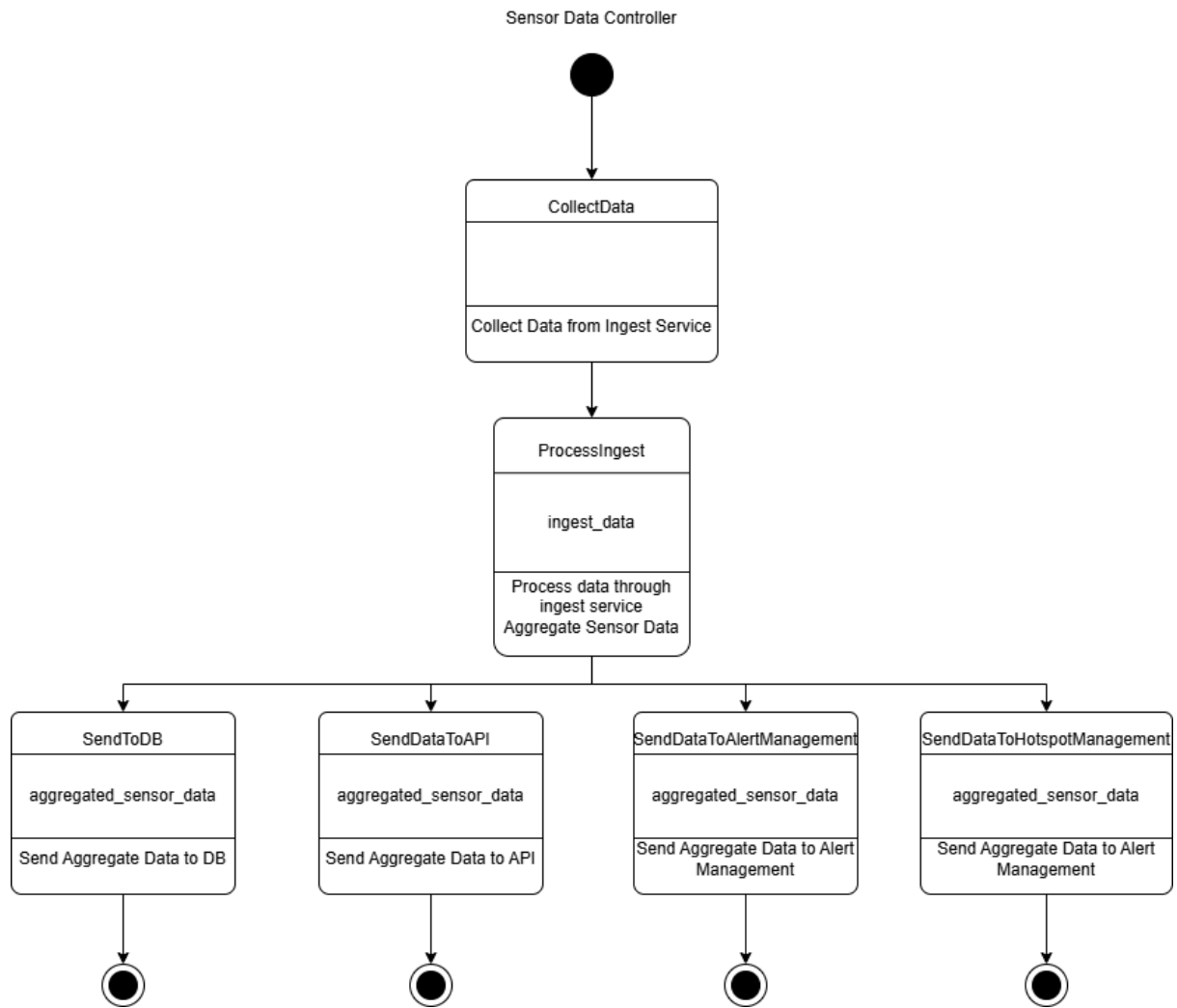


Figure 2.8. Sensor Data Controller State Diagram

3 Sequence Diagrams

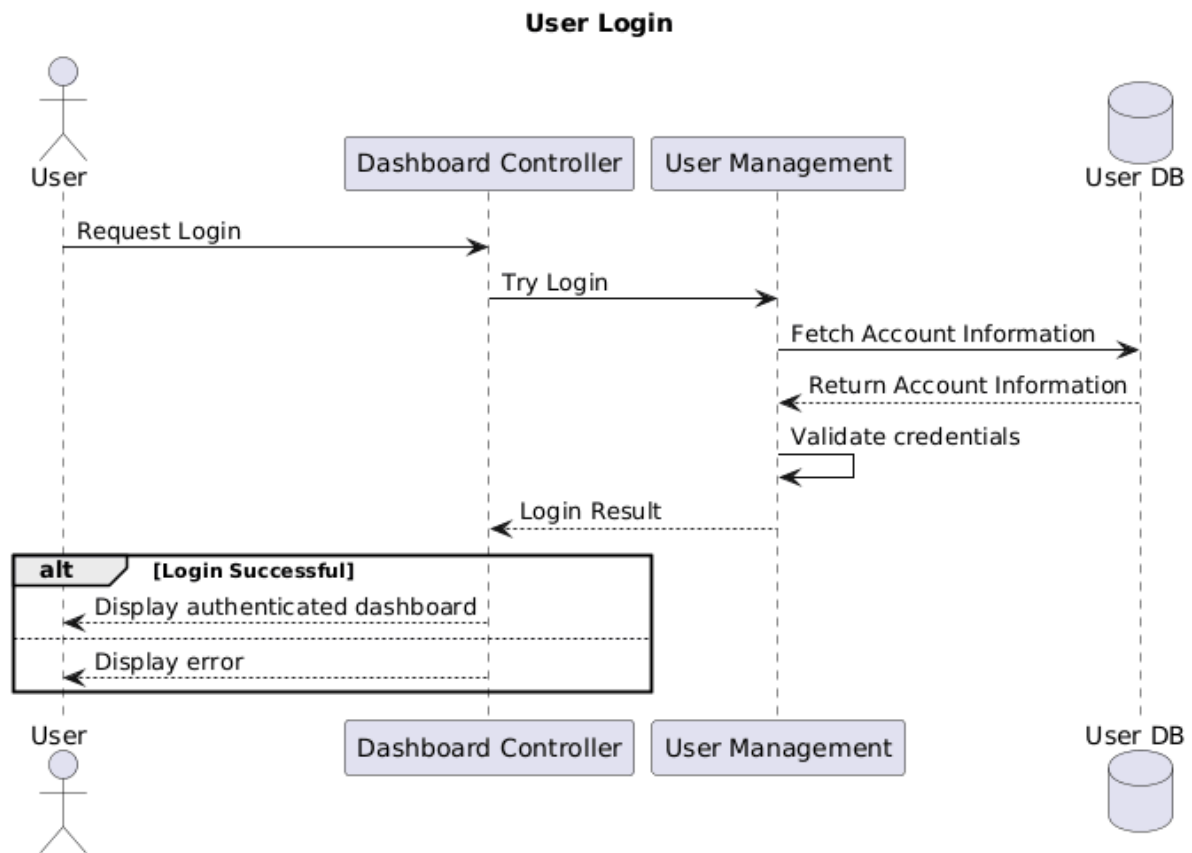


Figure 3.1. User Login Sequence Diagram

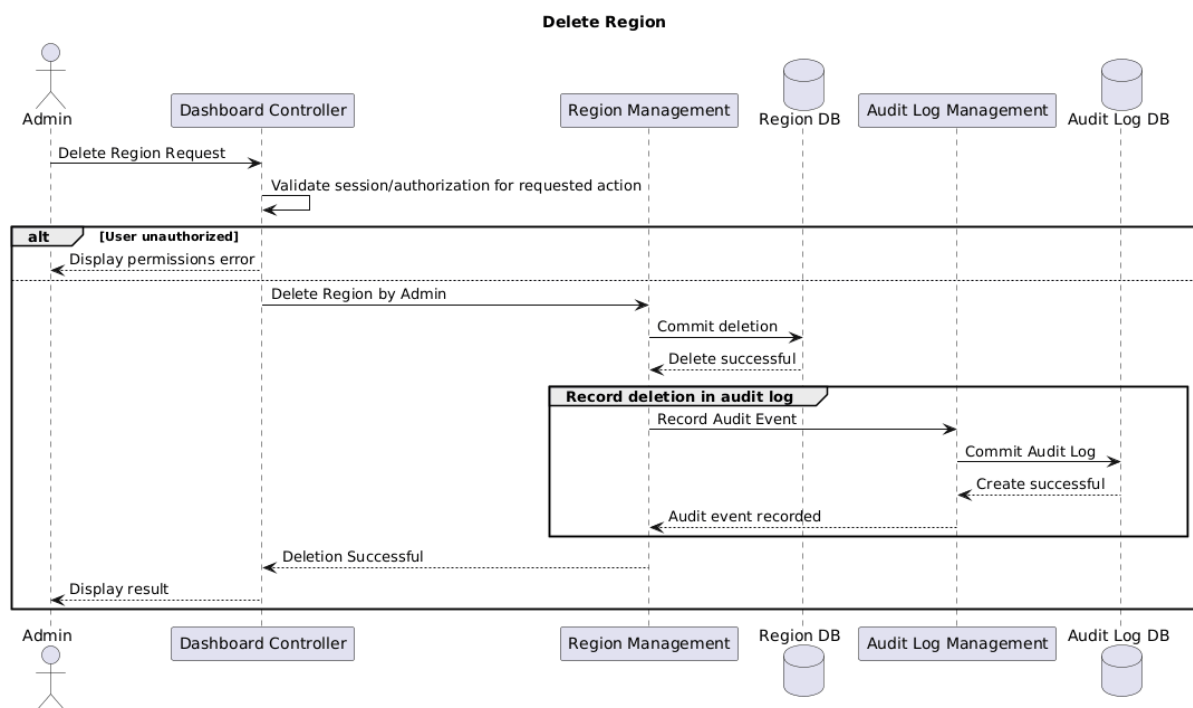


Figure 3.2. Delete Region Sequence Diagram

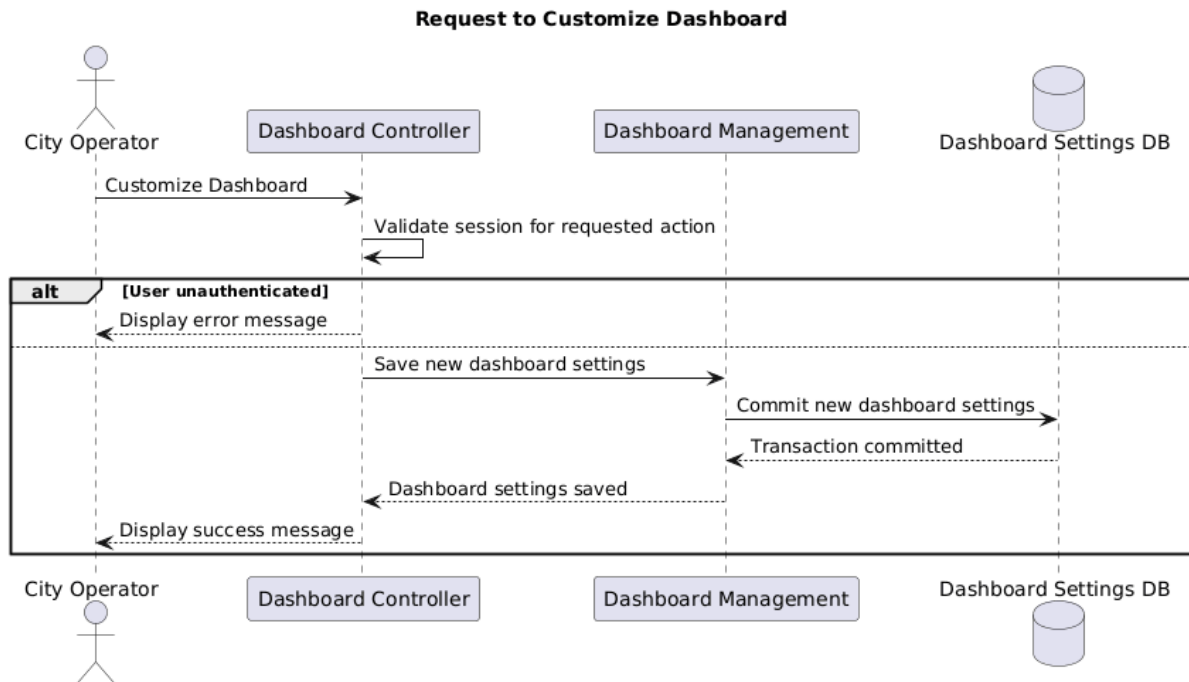


Figure 3.3. Dashboard Customization Sequence Diagram

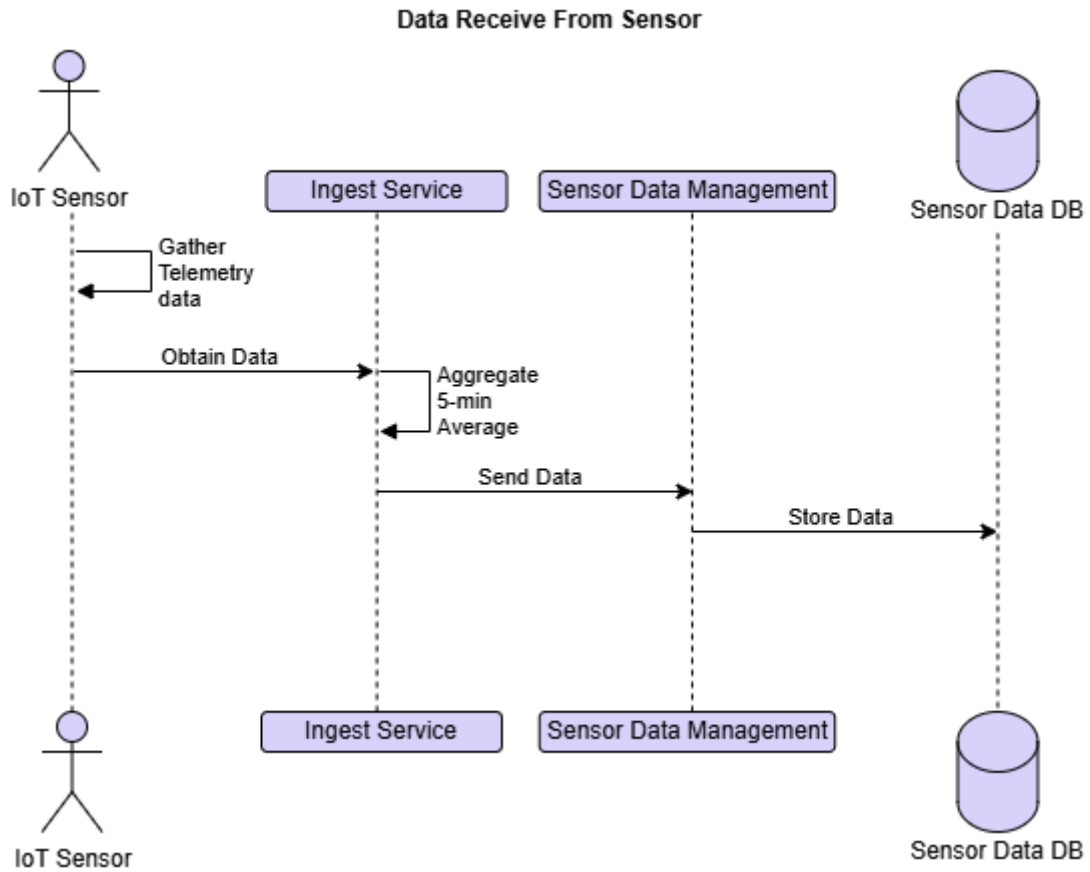


Figure 3.4. Data Receive From Sensor Sequence Diagram

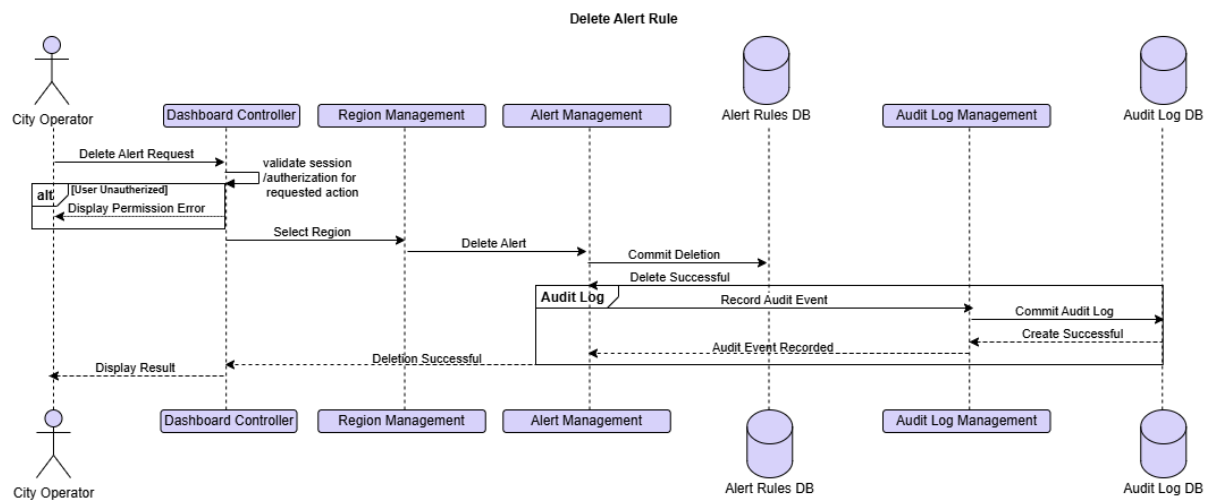


Figure 3.5. Delete Alert Rule Sequence Diagram

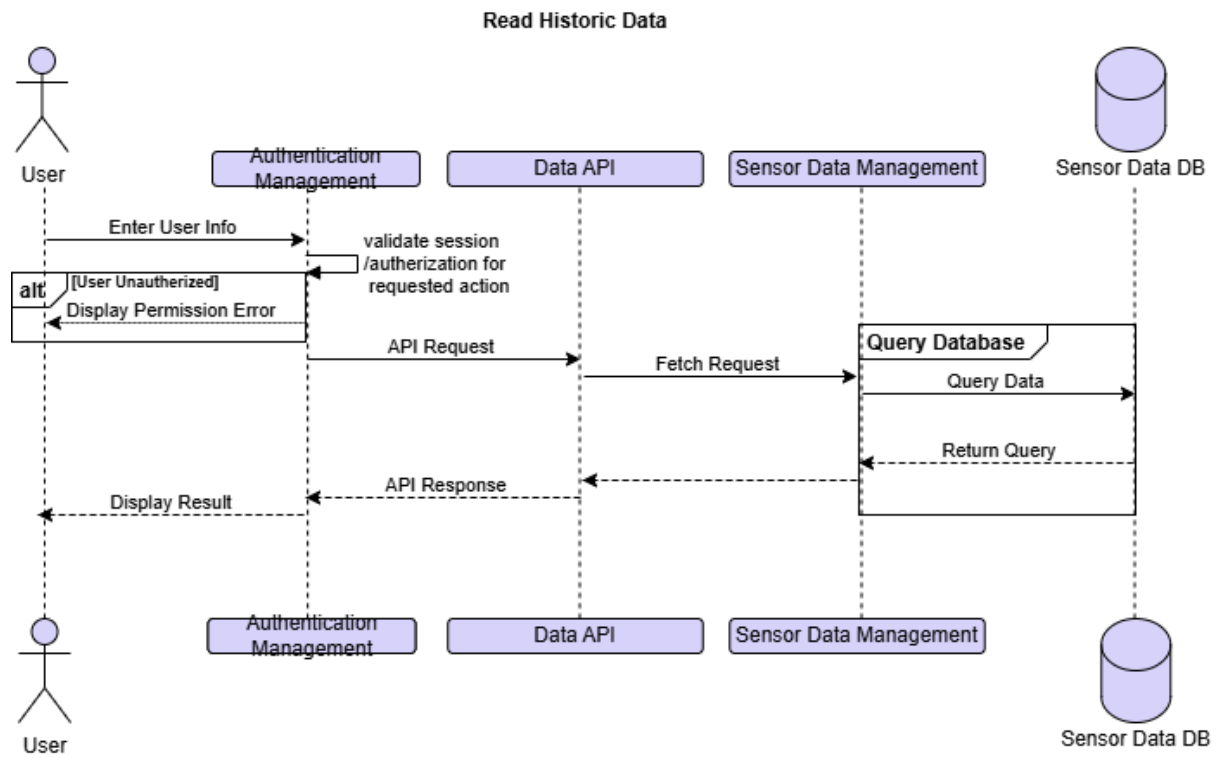


Figure 3.6. Read Historic Data Sequence Diagram

4 Detailed Class Diagram

This section should provide a detailed class diagram for your application.

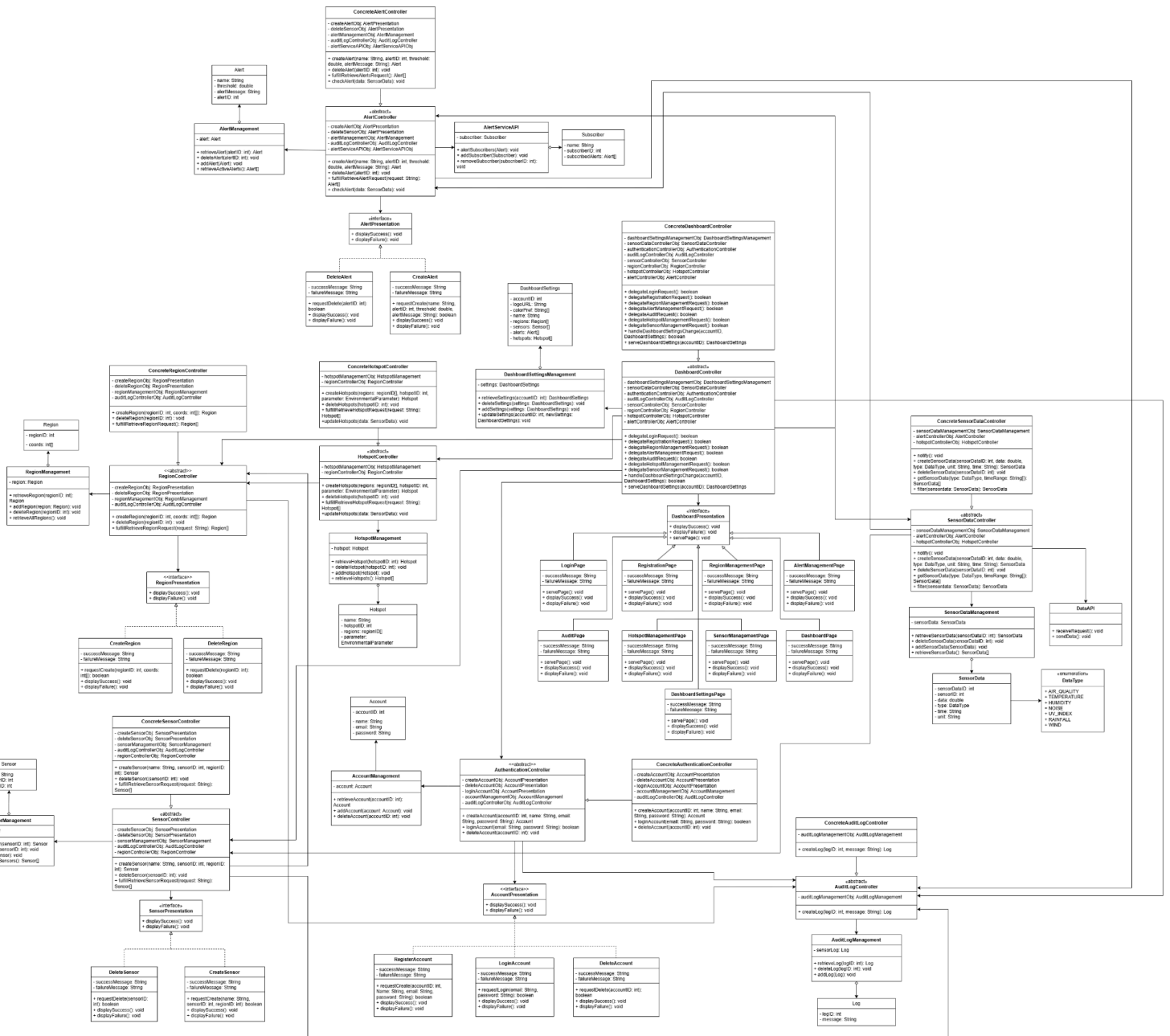


Figure 4.1. Overview of Detailed Class Diagram

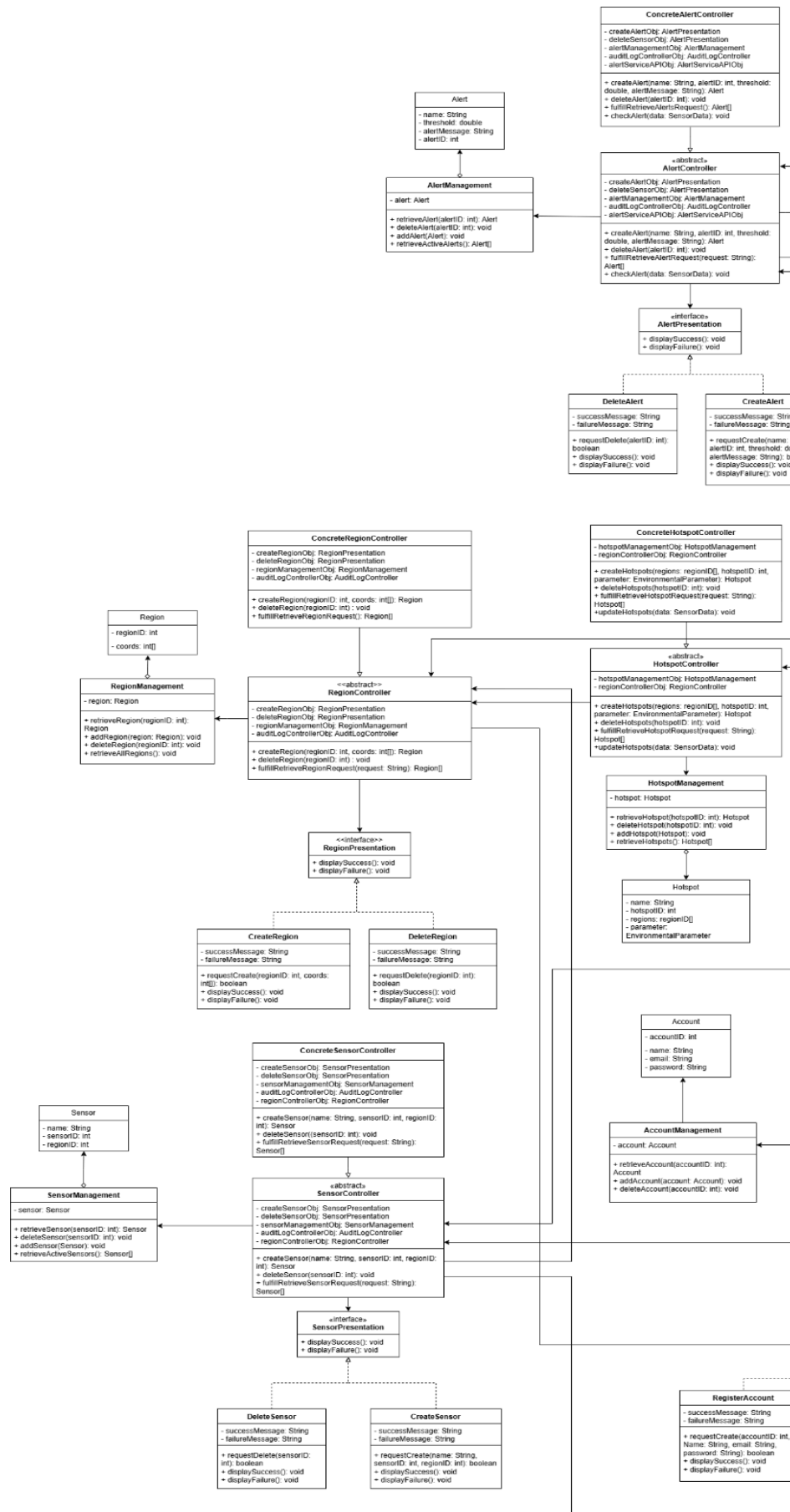


Figure 4.2. Left Third of Detailed Class Diagram

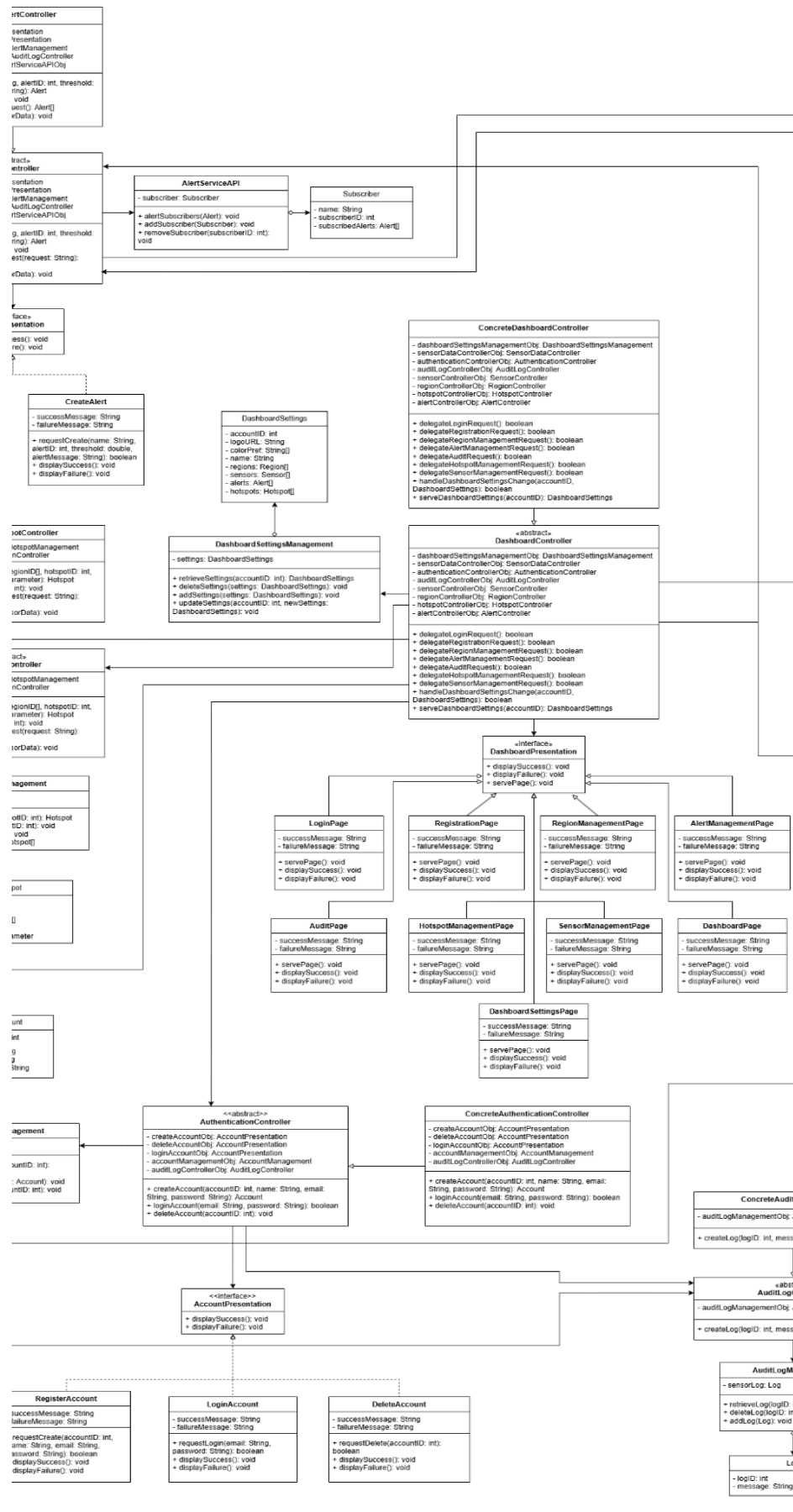


Figure 4.2. Middle Third of Detailed Class Diagram

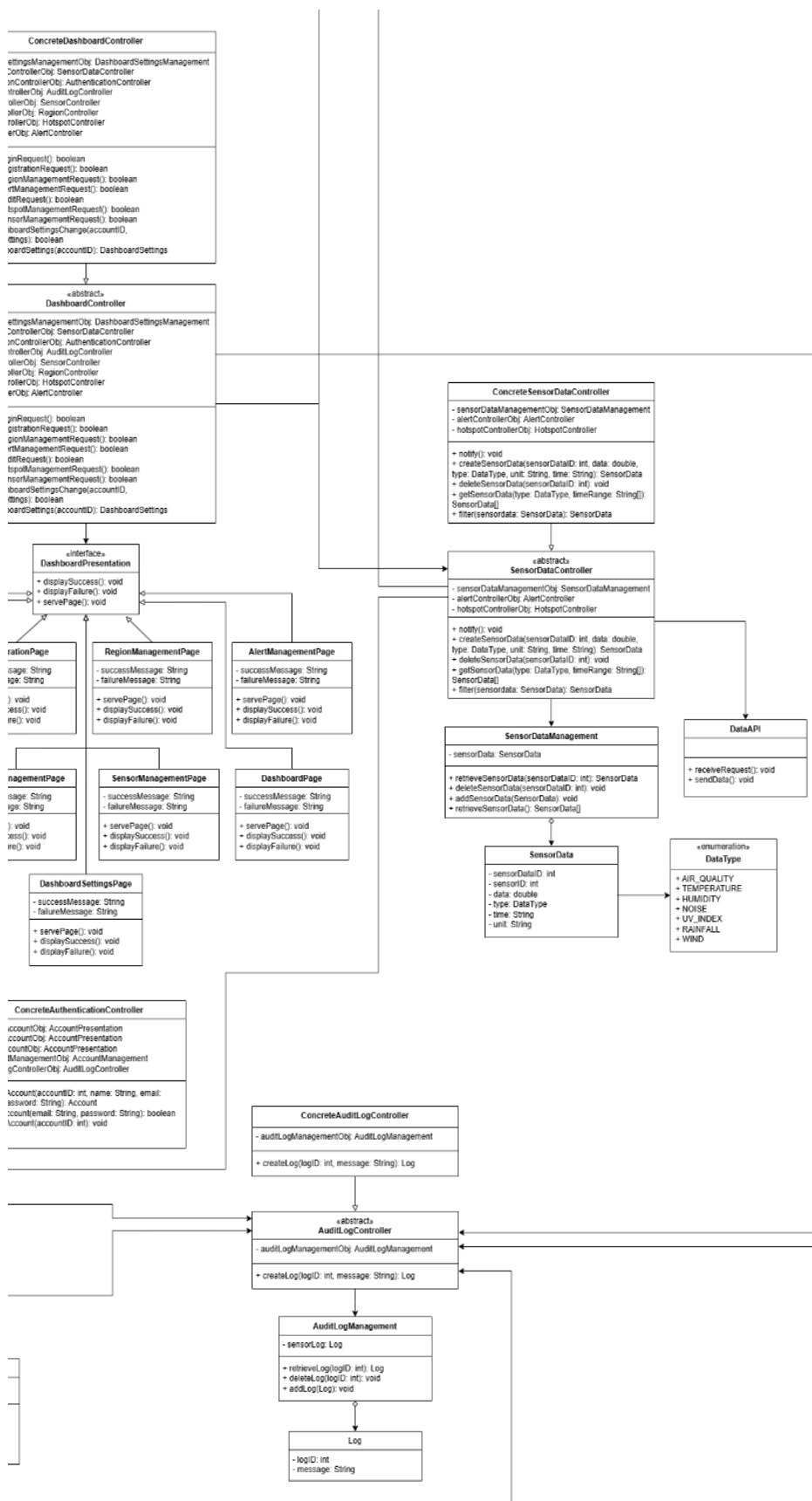


Figure 4.3. Right Third of Detailed Class Diagram

A Division of Labour

Include a Division of Labour sheet which indicates the contributions of each team member. This sheet must be signed by all team members.

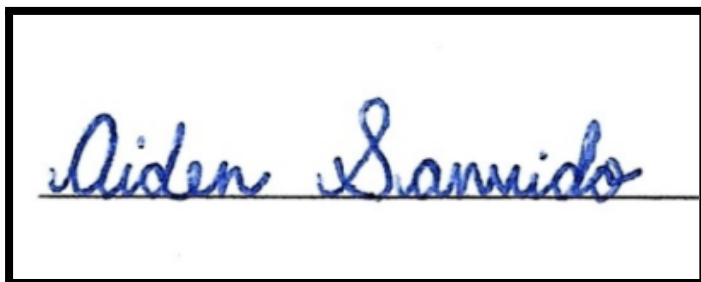
Kurlan Beeharry

- Section 4, Detailed Class Diagram
 - Region Agent
 - Sensor Data Agent
 - Authentication Agent
 - Co-created Hotspot Agent
 - Co-created Top Level Dashboard Agent



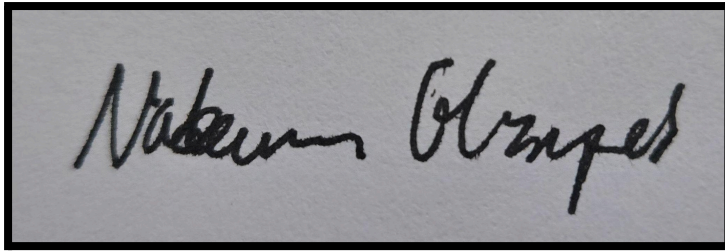
Aiden Sanvido

- Section 4, Detailed Class Diagram
 - Sensor Agent
 - Alert Agent
 - Audit Log Agent
 - Co-created Top Level Dashboard Agent
 - Co-created Hotspot Agent



Nadeem Elsayed

- Section 2, State Charts for Controller Classes
- Pasted Section 1 from previous deliverables

A rectangular box containing a handwritten signature in black ink. The signature is written in a cursive style and appears to read "Muhammad Ullah".

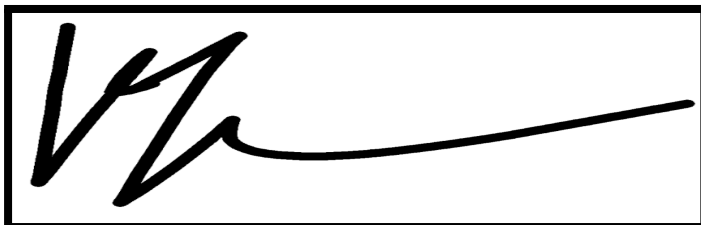
Muhammad Ullah

- Section 3, Sequence Diagrams
 - Data Received From Sensor
 - Delete Alert Rule
 - Read Historic Data

A rectangular box containing a handwritten signature in black ink. The signature is written in a cursive style and appears to read "Kaitlyn Kenwell".

Kaitlyn Kenwell

- Section 3, Sequence Diagrams
 - User Login
 - Delete Region
 - Dashboard Customization

A rectangular box containing a handwritten signature in black ink. The signature is written in a cursive style and appears to read "Muhammad Ullah".